

REPORT

BFS (Breadth-First Search)

Aim

To implement Breadth-First Search (BFS) to traverse a graph by visiting vertices level by level using a queue.

Objectives

- To understand the concept of graph traversal.
- To implement BFS using a queue.
- To visit all reachable vertices systematically.
- To understand the FIFO principle of a queue.

Requirements

- Computer/Laptop
- Programming language such as C, C++, Java, or Python
- Code editor/IDE
- Graph with vertices and edges

Theory

Breadth-First Search (BFS) is a graph traversal technique that visits vertices level by level. It starts from a selected vertex and first visits all its adjacent vertices before moving to the next level. BFS uses a **Queue**, which follows the **FIFO (First In, First Out)** principle. A visited list is maintained to avoid visiting the same vertex repeatedly.

Algorithm

1. Start with the given starting vertex.
2. Mark the starting vertex as visited.
3. Insert the starting vertex into the queue.
4. Remove the front vertex from the queue.
5. Check all adjacent vertices of the removed vertex.
6. Mark each unvisited adjacent vertex as visited.

7. Insert the newly visited vertices into the queue.
8. Repeat steps 4–7 until the queue becomes empty.
9. Display the vertices in the order they were visited.

PROGRAM:

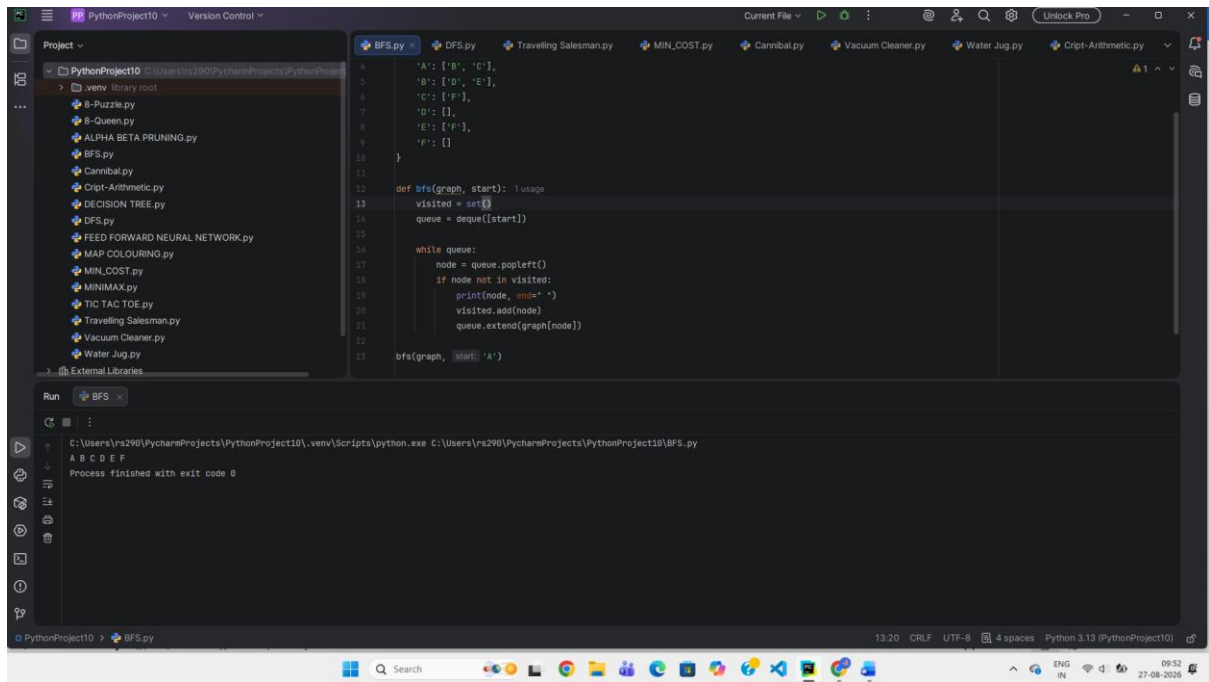
```
from collections import deque
```

```
graph = {  
    'A': ['B', 'C'],  
    'B': ['D', 'E'],  
    'C': ['F'],  
    'D': [],  
    'E': ['F'],  
    'F': []  
}
```

```
def bfs(graph, start):  
    visited = set()  
    queue = deque([start])  
  
    while queue:  
        node = queue.popleft()  
        if node not in visited:  
            print(node, end=" ")  
            visited.add(node)  
            queue.extend(graph[node])
```

```
bfs(graph, 'A')
```

OUTPUT:



The screenshot shows a Python IDE with a project named 'PythonProject10'. The left sidebar displays a file explorer with various Python files, including 'BFS.py'. The main editor window shows the code for 'BFS.py', which defines a graph and a breadth-first search function. The graph has nodes 'A' through 'F' with their respective neighbors. The BFS function starts at node 'A' and traverses the graph level by level. The bottom panel shows the output of the program, which prints the nodes in the order they were visited: 'A B C D E F'. The process finished with exit code 0.

```
graph LR
    A --- B
    A --- C
    B --- D
    C --- E
    D --- F
    E --- F
```

```
def bfs(graph, start):
    visited = set()
    queue = deque([start])

    while queue:
        node = queue.popleft()
        if node not in visited:
            print(node, end=" ")
            visited.add(node)
            queue.extend(graph[node])

    bfs(graph, start='A')
```

Run Output:

```
C:\Users\r290\PycharmProjects\PythonProject10\.venv\Scripts\python.exe C:\Users\r290\PycharmProjects\PythonProject10\BFS.py
A B C D E F
Process finished with exit code 0
```

Result

Thus, the **Breadth-First Search (BFS)** traversal of the given graph is successfully performed using a queue, visiting the vertices level by level.